

All Correlations Yield the Same Peak Work, But Quantum is a Better Fuel

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Initial system–environment correlations are a thermodynamic resource, enabling work extraction via their erasure. We compare the work potential of classical, quantum, and hypothetical stronger-than-quantum correlations as a function of measurement misalignment. While all models can yield a peak extractable work of $k_B T \ln 2$, corresponding to a mutual information of $\ln 2$ (the maximum for a single unbiased binary outcome), their value as a resource differs critically in robustness. The classical resource is fragile, with its mutual information degrading as $\Theta(\delta\theta \ln(1/\delta\theta))$ under a misalignment $\delta\theta$, whereas the quantum resource is robust, degrading only as $\Theta(\delta\theta^2 \ln(1/\delta\theta))$. One stronger-than-quantum resource is perfectly robust for all but a single critical angle. This hierarchy in operational robustness is quantitatively expressed through an aggregate thermodynamic robustness parameter evaluated across the measurement settings of a CHSH-type Bell test.

I. INTRODUCTION

Classical thermodynamics defines an isolated system by boundaries impermeable to matter and energy. This picture is challenged by microscopic systems prepared with initial correlations to their environment. Even when sealed against energy exchange, such a system’s local state can be altered by operations on its correlated environmental counterpart. An agent possessing side information can consume these correlations to perform work, a process governed by generalized thermodynamic laws that incorporate information-theoretic balances [1–3].

The connection between information and thermodynamics has a long history, originating with Maxwell’s “demon” thought experiment and formalized through the work of Szilard [4], who showed that a single bit of information about a molecule’s position in a partitioned box can be converted into $k_B T \ln 2$ of work. Landauer [5] later established that this cycle is closed by the erasure cost: resetting the demon’s memory dissipates at least $k_B T \ln 2$ of heat, thereby saving the second law [6, 7].

In the quantum regime, information-thermodynamic trade-offs are enriched by genuinely non-classical correlations. It has been shown that quantum discord [8, 9] and quantum entanglement [2] can serve as thermodynamic resources, and that the framework of generalized free energies [10–12] extends the second law to the single-shot regime. Sagawa and Ueda [13, 14] established rigorous equalities connecting feedback control, measurement, and work extraction in the quantum setting. Comprehensive reviews of the field can be found in Refs. [15, 16].

This paper demonstrates that the *nonlocal character* of a correlation law maps directly to its *operational value* as a thermodynamic resource. We show that while classical, quantum, and stronger-than-quantum correlations share the same peak energetic value, they exhibit a clear hierarchy in their robustness against experimental imperfection. This robustness hierarchy is quantified by an ag-

gregate thermodynamic parameter inspired by the measurement settings of the Clauser–Horne–Shimony–Holt (CHSH) scenario [17–19], though we emphasize that this parameter is an operational figure of merit, not a Bell inequality.

The remainder of this paper is organized as follows. Section II reviews the information-theoretic framework connecting mutual information to extractable work and derives the mutual information for binary-outcome correlations. Section III describes the operational work-extraction protocol. Section IV introduces the three representative correlation laws. Section V analyzes their thermodynamic value and robustness, establishing a general smoothness criterion. Section VI defines and evaluates an aggregate thermodynamic robustness parameter across multiple measurement settings. Section VII discusses extensions and limitations, including the relaxation of the binary-outcome and uniform-marginal assumptions. In simple terms, a *uniform marginal* means that if one looks at the results of just one party (“Alice alone” or “Bob alone”) and ignores the other, their outcomes appear completely random and equally distributed—the output is indistinguishable from pure noise. It appears as a sequence of perfectly random coin flips (+1 or −1 with 50% probability) that reveals absolutely no information on its own; the meaningful structure exists solely in the relationship between the two sides. Section VIII positions the work relative to the existing literature. Section IX concludes.

II. PRELIMINARIES

A. From information to work

Consider a bipartite system consisting of a subsystem A and its environment E , jointly described by a state ρ_{AE} with reduced states $\rho_A = \text{Tr}_E[\rho_{AE}]$ and $\rho_E = \text{Tr}_A[\rho_{AE}]$. The quantum mutual information,

$$I(A : E) = S(\rho_A) + S(\rho_E) - S(\rho_{AE}), \quad (1)$$

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where $S(\rho) = -\text{Tr}[\rho \ln \rho]$ is the von Neumann entropy, quantifies the total correlations (both classical and quantum) between A and E .

If A is in contact with a thermal bath at temperature T , and the agent performs a cyclic process on A that begins with correlations $I_i(A:E)$ and terminates in an uncorrelated product state $\rho_A \otimes \rho_E$, the generalized second law [1] imposes the following bound on the extractable work:

$$W_{\text{ext}} \leq -\Delta F_T(\rho_A) + k_B T I_i(A:E), \quad (2)$$

where $\Delta F_T(\rho_A)$ is the change in the non-equilibrium free energy of subsystem A . This result can be understood intuitively: the mutual information $I_i(A:E)$ represents a “bonus” resource, additional to the system’s own free energy, that an informed agent can exploit. When the process is cyclic on A , so that $\Delta F_T(\rho_A) = 0$, this simplifies to

$$W_{\text{ext}} \leq k_B T I_i(A:E). \quad (3)$$

The physical content of Eq. (3) generalizes Szilard’s result [4]: one bit of mutual information ($I = \ln 2$ nats) can be reversibly converted into $k_B T \ln 2$ of work, precisely the energy liberated by an isothermal expansion of a single-molecule gas when the piston’s initial position is known. Following del Rio *et al.* [2], the work cost of erasing a system is governed by the conditional entropy $S(A|E) = S(\rho_{AE}) - S(\rho_E)$, which can be negative for entangled states, permitting work *gain* during erasure. The work bound in Eq. (3) is consistent with this picture, since for a maximally mixed marginal ρ_A (entropy $S(\rho_A) = \ln 2$), we have $I(A:E) = \ln 2 - S(A|E)$.

Scope of the bound. The quantum mutual information of Eq. (1) can exceed $\ln 2$ for quantum subsystems (for instance, $I(A:E) = 2 \ln 2$ for a maximally entangled two-qubit state), and in principle more general protocols could exploit such larger correlations [2]. In the present work we restrict attention to the *outcome-level* mutual information arising from binary measurements with uniform marginals, for which $I \leq \ln 2$ always holds. The bound $W \leq k_B T \ln 2$ is therefore specific to this one-unbiased-bit operational setting, not universal across all quantum thermodynamic protocols.

B. Mutual information from correlations

We now specialize to a scenario directly relevant to Bell-type experiments. Consider two parties sharing a no-signaling bipartite system with binary outcomes $a, e \in \{+1, -1\}$ and uniform marginals: $P(a) = P(e) = 1/2$. In this case, the joint distribution is entirely determined by the correlation function

$$E(\theta) = \langle a e \rangle_\theta = \sum_{a,e} a e P(a, e|\theta), \quad (4)$$

where θ is the relative angle between the two measurement settings. Explicitly, the joint probabilities are

$$P(a = e|\theta) = \frac{1 + E(\theta)}{2}, \quad P(a \neq e|\theta) = \frac{1 - E(\theta)}{2}. \quad (5)$$

The marginal entropy is $H(A) = \ln 2$. The conditional entropy is the binary entropy of the error probability,

$$H(A|E) = h_2\left(\frac{1 - E(\theta)}{2}\right) = h_2\left(\frac{1 + E(\theta)}{2}\right), \quad (6)$$

where $h_2(p) = -p \ln p - (1-p) \ln(1-p)$ is the binary Shannon entropy in nats, and the last equality follows from the symmetry $h_2(p) = h_2(1-p)$. The mutual information is therefore

$$I_\theta(A:E) = H(A) - H(A|E) = \ln 2 - h_2\left(\frac{1 + E(\theta)}{2}\right). \quad (7)$$

This result depends on the correlation $E(\theta)$ *only through its absolute value* $|E(\theta)|$, via the binary entropy. Perfect correlation or anti-correlation ($|E| = 1$) gives $I = \ln 2$; zero correlation ($E = 0$) gives $I = 0$. Note that Eq. (7) is *model-independent*: it holds for any bipartite device (or theory) whose outputs are binary with uniform marginals, in which case the joint distribution is fully determined by the correlator $E(\theta) = \langle AE \rangle_\theta$.

III. OPERATIONAL PROTOCOL

To make the work-extraction scenario concrete, we describe an operational protocol inspired by the Szilard engine [4, 13]. We treat the measurement outcomes as classical registers; hence $I_\theta(A:E)$ denotes the *Shannon* mutual information of the binary variables $A, E \in \{\pm 1\}$.

1. Alice and Bob share a correlated resource (classical shared randomness, an entangled quantum state, or a no-signaling box).
2. A referee assigns measurement settings: Alice receives setting x (direction α_x) and Bob receives setting y (direction β_y). The settings (equivalently, the relative angle $\theta = \alpha_x - \beta_y$) are assumed to be known to both parties (e.g., publicly announced by the referee).
3. Alice performs her measurement and obtains $a \in \{+1, -1\}$. Bob queries his device with setting y , obtaining $e \in \{+1, -1\}$, which is stored in a physical two-state working medium (e.g., Left/Right of a one-molecule Szilard box at temperature T).
4. Alice communicates her one-bit result a to Bob via a classical channel.
5. Using a and the known correlator $E(\theta) = \langle AE \rangle_\theta$, Bob performs a reversible relabeling (a conditional

flip) of his two-state working medium and defines an “agreement” variable

$$z := \sigma(\theta) a e \in \{+1, -1\},$$

where $\sigma(\theta) \in \{+1, -1\}$ is chosen to maximize $P(z = +1|\theta)$ (equivalently, $\sigma(\theta) = \text{sgn } E(\theta)$ whenever $E(\theta) \neq 0$). For uniform marginals,

$$\begin{aligned} p_{\text{corr}}(\theta) &= P(z = +1|\theta) \\ &= \max\left\{\frac{1 + E(\theta)}{2}, \frac{1 - E(\theta)}{2}\right\} \\ &= \frac{1 + |E(\theta)|}{2}. \end{aligned} \quad (8)$$

6. Bob now performs a feedback-controlled quasi-static isothermal protocol that brings his two-state working medium from the biased distribution $P(z)$ to thermal equilibrium (uniform). The A – E correlation is the informational resource being consumed. The maximum average extractable work per run is [13, 20]

$$W(\theta) = k_B T I_\theta(A:E) = k_B T \left[\ln 2 - h_2\left(\frac{1 + |E(\theta)|}{2}\right) \right]. \quad (9)$$

Since Alice communicates exactly one classical bit and the marginals are uniform, the mutual information satisfies $I_\theta(A:E) \leq H(A) = \ln 2$, hence $W(\theta) \leq k_B T \ln 2$ in this one-unbiased-bit setting [5, 6]. This bound is saturated when $|E(\theta)| = 1$, i.e., when Bob can deterministically predict (equivalently, deterministically relabel) the logical state of his two-state working medium.

IV. A HIERARCHY OF CORRELATION LAWS

We analyze three representative bipartite correlation laws $E(\theta)$ for $\{\pm 1\}$ -outcome measurements as a function of the relative angle $\theta \in [0, \pi]$ between settings. Their nonlocality is gauged by the CHSH parameter [17, 18]

$$\mathcal{S}_{\text{CHSH}} = |E(\theta_1, \theta_2) + E(\theta_1, \theta'_2) + E(\theta'_1, \theta_2) - E(\theta'_1, \theta'_2)|, \quad (10)$$

for which local realism imposes a bound of 2, quantum mechanics a bound of $2\sqrt{2}$ (the Tsirelson bound [21]), and the no-signaling principle an algebraic limit of 4. The three models are (see Fig. 1):

1. **Classical (linear / “triangle” model):**

$$E_c(\theta) = -1 + \frac{2\theta}{\pi} \quad (0 \leq \theta \leq \pi),$$

with the extension to other angles understood by the usual symmetries. This is a standard *rotationally covariant* local-hidden-variable (LHV) correlation satisfying the singlet boundary condition

$E(0) = -1$ (perfect anti-correlation at zero misalignment) and unbiased marginals. For the conventional CHSH choice of settings used in this work, it attains the local bound $\mathcal{S}_{\text{CHSH}}^{(c)} = 2$. [22]

2. **Quantum (cosine):** $E_q(\theta) = -\cos \theta$. This describes measurements on a two-qubit singlet state and reaches the Tsirelson bound, $\mathcal{S}_{\text{CHSH}}^{(q)} = 2\sqrt{2}$.

3. **Stronger-than-quantum (step):** $E_s(\theta) = \text{sgn}(2\theta/\pi - 1)$. This hypothetical “box” correlation saturates the algebraic limit, $\mathcal{S}_{\text{CHSH}}^{(s)} = 4$ [23, 24]. We note that the standard Popescu–Rohrlich box is defined for finitely many settings; the extension to a continuous function $E(\theta)$ is an additional idealization consistent with no-signaling but not uniquely fixed by the PR-box definition.

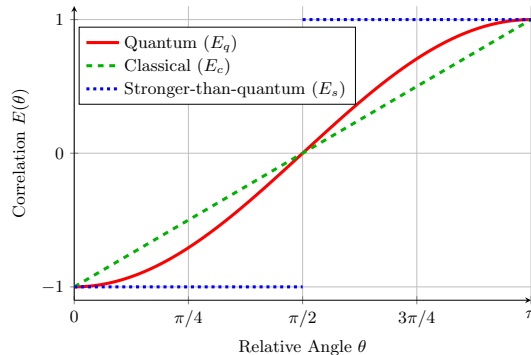


FIG. 1. Comparison of the three correlation functions $E(\theta)$. The quantum (solid red) and classical (dashed green) correlations only achieve perfect anti-correlation at $\theta = 0$. The stronger-than-quantum correlation (dotted blue) is perfect for all angles except $\theta = \pi/2$.

V. THERMODYNAMIC VALUE AND ROBUSTNESS

A. Peak work and mutual information

The thermodynamic value of these correlations is governed by the mutual information $I_\theta(A:E)$ derived in Eq. (7). As shown in Fig. 2, perfect correlation or anti-correlation ($|E(\theta)| = 1$) yields the maximal mutual information $I = \ln 2$. Consequently, according to Eq. (3), the maximum work extractable per bipartite system in this binary-outcome, uniform-marginal setting is bounded by $k_B T \ln 2$. This peak value is achievable in all three models, for instance, by aligning the measurement bases ($\theta = 0$).

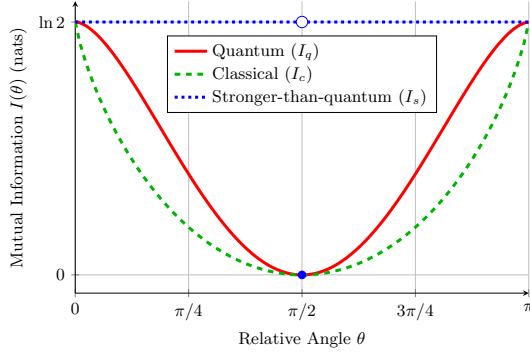


FIG. 2. Mutual information versus measurement angle θ . All models achieve the maximum value of $\ln 2$, but their robustness to misalignment differs. Classical information (dashed green) degrades as $\Theta(\delta\theta \ln(1/\delta\theta))$ from the peak, while quantum (solid red) degrades as $\Theta(\delta\theta^2 \ln(1/\delta\theta))$, offering superior resilience.

B. Robustness analysis

This thermodynamic equivalence in peak yield belies a critical operational difference: robustness. The practical advantage of non-classical correlations becomes apparent when considering misalignment from the optimal setting ($\theta = 0$).

a. Classical (linear) robustness. For the classical model, $E_c(\delta\theta) = -1 + 2\delta\theta/\pi$ near $\theta = 0$. Setting $p = (1 + E_c)/2 = \delta\theta/\pi$, the binary entropy for small p behaves as $h_2(p) \approx -p \ln p + p + \mathcal{O}(p^2)$. Therefore,

$$I_c(\delta\theta) = \ln 2 - h_2\left(\frac{\delta\theta}{\pi}\right) \approx \ln 2 - \frac{\delta\theta}{\pi} \ln \frac{\pi}{\delta\theta} - \frac{\delta\theta}{\pi} + \dots \quad (11)$$

The leading correction is $\Theta(\delta\theta \ln(1/\delta\theta))$ —worse than linear: the mutual information degrades sharply for any nonzero misalignment. This makes the classical resource *brittle* with respect to experimental noise.

b. Quantum (cosine) robustness. For the quantum model, $E_q(\delta\theta) = -\cos \delta\theta \approx -1 + \delta\theta^2/2$. Setting $p = (1 + E_q)/2 \approx \delta\theta^2/4$, we have $h_2(p) \approx -p \ln p + p \approx \frac{\delta\theta^2}{4} \ln \frac{4}{\delta\theta^2} + \frac{\delta\theta^2}{4}$ for small $\delta\theta$. More precisely, expanding Eq. (7) with $E_q = -\cos \theta$ we can write

$$I_q(\delta\theta) = \ln 2 - h_2\left(\sin^2 \frac{\delta\theta}{2}\right) \approx \ln 2 - \Theta(\delta\theta^2 \ln(1/\delta\theta)). \quad (12)$$

The leading correction is *quadratic* (up to logarithmic factors). The quantum correlation $E_q(\theta) = -\cos \theta$ has a *stationary point* at $\theta = 0$ (i.e., $E'_q(0) = \sin 0 = 0$), which is the fundamental reason for this improved robustness. The quantum resource is significantly more *resilient* against small imperfections in establishing a shared reference frame.

c. Stronger-than-quantum (step) robustness. The stronger-than-quantum model provides the ultimate in robustness: for any $\delta\theta \neq \pi/2$, we have $|E_s(\delta\theta)| = 1$ and

therefore $I_s(\delta\theta) = \ln 2$. Coordination of measurement settings is almost entirely unnecessary; the resource is maximal for all but a single critical angle.

d. General criterion. The hierarchy above is not an accident of the three specific models: it reflects a general principle. For *any* correlation function $E(\theta)$ satisfying $E(0) = -1$, the robustness of the mutual information I_θ near $\theta = 0$ is determined by the *order of the leading nonzero term in the expansion of $1 + E(\theta)$ around the extremum*:

1. If $E(\theta) = -1 + c\theta + o(\theta)$ with $c > 0$ (nonzero first-order coefficient, as for the linear model), $I(\delta\theta)$ drops away from $\ln 2$ at a rate $\Theta(\delta\theta \ln(1/\delta\theta))$, worse than linear.
2. If $E(\theta) = -1 + c\theta^2 + o(\theta^2)$ with $c > 0$ (vanishing first-order coefficient, i.e., a stationary point as for the cosine correlation), the mutual information decays only quadratically (up to logarithmic factors): $\Theta(\delta\theta^2 \ln(1/\delta\theta))$. This applies to any quantum state whose correlation function has a smooth extremum.
3. If $|E(\theta)| = 1$ is maintained over a finite neighborhood of $\theta = 0$, the resource is locally flat and perfectly robust.

More generally, if $E(\theta) = -1 + c|\theta|^n + o(|\theta|^n)$ with $c > 0$ and $n > 0$, the degradation scales as $\Theta(|\theta|^n \ln(1/|\theta|))$. Thus, the advantage of quantum correlations in this thermodynamic setting is not tied to the singlet state alone but is generic for any quantum correlation function *whose first-order coefficient vanishes at its extremum*—a direct consequence of the smoothness guaranteed by the Born rule and unitary evolution.

Table I summarizes the key properties of the three correlation classes.

VI. MULTI-SETTING THERMODYNAMIC ROBUSTNESS

The operational hierarchy established above can be quantified by an aggregate robustness parameter that evaluates the total thermodynamic value across multiple measurement settings, as would arise in a protocol where the measurement angle varies across runs.

We adopt the standard CHSH measurement configuration—Alice’s settings $\alpha_1 = 0$, $\alpha_2 = \pi/2$ and Bob’s settings $\beta_1 = \pi/4$, $\beta_2 = -\pi/4$ —which produces four setting pairs with relative angles $\theta_{11} = \pi/4$, $\theta_{12} = \pi/4$, $\theta_{21} = \pi/4$, $\theta_{22} = 3\pi/4$. For isotropic correlations satisfying $|E(\theta)| = |E(\pi - \theta)|$, one has $I(\pi/4) = I(3\pi/4)$, so the four terms all take the same value.

We define the aggregate thermodynamic robustness parameter

$$\mathcal{S}_W = k_B T \sum_{k=1}^4 I(\theta_k) = k_B T [3I(\pi/4) + I(3\pi/4)]. \quad (13)$$

TABLE I. Summary of the three correlation classes. $\mathcal{S}_{\text{CHSH}}$ is the CHSH value; W_{max} is the peak extractable work (in units of $k_B T \ln 2$); the “Degradation of I ” column gives the leading magnitude of the correction to $I(\delta\theta) = \ln 2 - \dots$ for small misalignment $\delta\theta$.

Class	$\mathcal{S}_{\text{CHSH}}$	$W_{\text{max}}/k_B T \ln 2$	Degradation of I near $\theta=0$
Classical	2	1	$\Theta(\delta\theta \ln(1/\delta\theta))$
Quantum	$2\sqrt{2}$	1	$\Theta(\delta\theta^2 \ln(1/\delta\theta))$
Stronger-than-quantum	4	1	0 (perfectly robust)

Important remark. Although $\mathcal{S}_W$ is evaluated at the CHSH measurement angles, it is *not* a Bell inequality: no local bound is derived or claimed. The standard CHSH functional depends on the *signs* of the correlators through the combination $E+E+E-E$, which is what enables the separation between local and nonlocal theories. The mutual information $I(\theta)$ depends only on $|E(\theta)|$ and is therefore insensitive to the sign structure. We use $\mathcal{S}_W$ purely as an operational figure of merit quantifying how much total work can be extracted when measurement settings span the CHSH configuration. The hierarchy $\mathcal{S}_W^{(c)} < \mathcal{S}_W^{(q)} < \mathcal{S}_W^{(s)}$ reflects the robustness advantage of non-classical correlations: stronger correlations maintain a higher $|E|$ (and hence higher I) across diverse measurement angles.

We now evaluate $\mathcal{S}_W$ explicitly for each model.

a. *Classical:* Using Eq. (7) with $E_c(\pi/4) = -1/2$ and $E_c(3\pi/4) = 1/2$, so that $|E| = 1/2$ in both cases, we obtain $I(\pi/4) = I(3\pi/4) = \ln 2 - h_2(1/4)$. Computing $h_2(1/4)$ explicitly:

$$\begin{aligned} h_2\left(\frac{1}{4}\right) &= -\frac{1}{4} \ln \frac{1}{4} - \frac{3}{4} \ln \frac{3}{4} \\ &= 2 \ln 2 - \frac{3}{4} \ln 3 \approx 0.5623 \text{ nats.} \end{aligned} \quad (14)$$

Therefore $I(\pi/4) = \ln 2 - (2 \ln 2 - \frac{3}{4} \ln 3) = \frac{3}{4} \ln 3 - \ln 2 \approx 0.1308$ nats, and

$$\mathcal{S}_W^{(c)} = 4 \times 0.1308 k_B T \approx 0.523 k_B T. \quad (15)$$

b. *Quantum:* With $E_q(\pi/4) = -\cos(\pi/4) = -1/\sqrt{2}$ and $E_q(3\pi/4) = -\cos(3\pi/4) = 1/\sqrt{2}$, we similarly have $|E| = 1/\sqrt{2}$ and $I(\pi/4) = I(3\pi/4) = \ln 2 - h_2(\frac{1+1/\sqrt{2}}{2})$, where $p = (1 - 1/\sqrt{2})/2 = \sin^2(\pi/8) \approx 0.1464$. Computing $h_2(0.1464)$ directly:

$$\begin{aligned} h_2(0.1464) &= -0.1464 \ln(0.1464) - 0.8536 \ln(0.8536) \\ &\approx 0.2813 + 0.1352 \\ &\approx 0.4165 \text{ nats.} \end{aligned} \quad (16)$$

Therefore $I(\pi/4) \approx 0.6931 - 0.4165 = 0.2766$ nats, and

$$\mathcal{S}_W^{(q)} = 4 \times 0.2766 k_B T \approx 1.107 k_B T. \quad (17)$$

c. *Stronger-than-quantum:* With $|E_s(\pi/4)| = |E_s(3\pi/4)| = 1$, we have $I(\pi/4) = I(3\pi/4) = \ln 2 \approx 0.6931$ nats. Therefore,

$$\mathcal{S}_W^{(s)} = 4 k_B T \ln 2 \approx 2.773 k_B T. \quad (18)$$

These results confirm the strict hierarchy

$$\mathcal{S}_W^{(c)} < \mathcal{S}_W^{(q)} < \mathcal{S}_W^{(s)}. \quad (19)$$

This hierarchy is a direct consequence of the robustness analysis of Sec. VB: stronger non-locality produces correlations whose magnitude $|E(\theta)|$ is maintained at larger misalignment angles, resulting in greater aggregate thermodynamic value across diverse measurement settings.

VII. EXTENSIONS AND LIMITATIONS

A. Relaxing the uniform-marginal assumption

The derivation of the mutual information in Eq. (7) relied on two structural assumptions: (i) binary outcomes and (ii) uniform marginals ($P(a) = P(e) = 1/2$). Assumption (ii) can be relaxed straightforwardly. For a general binary no-signaling distribution with marginals $P(a=+1) = q_A$ and $P(e=+1) = q_E$, the mutual information becomes

$$I(A:E) = H(A) + H(E) - H(A, E), \quad (20)$$

where all entropies are now Shannon entropies of the respective distributions. With non-uniform marginals, the joint distribution is no longer determined by $E(\theta)$ alone: one additionally needs the marginals. The work bound (3) remains valid, but the simple formula (7) is replaced by a more complex expression involving two parameters. For binary outcomes, the mutual information is bounded by $I(A:E) \leq \min\{H(A), H(E)\} \leq \ln 2$, so the peak extractable work remains at most $k_B T \ln 2$ regardless of the marginals. With non-uniform marginals, $\min\{H(A), H(E)\} < \ln 2$, so the achievable peak may be *lower* than in the uniform case.

Importantly, the *robustness hierarchy* is unaffected by this generalization. The key argument (Sec. VB, “General criterion”) depends only on the leading-order scaling of $1 + E(\theta)$ near its extremum, which is a property of the correlation law, not of the marginals. A smooth $E(\theta)$ produces a smooth $I(\theta)$, regardless of the marginal distribution, and the Taylor expansion around the peak retains the same leading order.

B. Beyond binary outcomes

For d -outcome measurements, the joint distribution is no longer captured by a single correlation function, and the conditional entropy involves the full $d \times d$ joint probability table. The extractable work is still bounded by $k_B T I(A:E)$, but the connection to a single “angle” becomes less direct. A natural generalization replaces $E(\theta)$ by a family of correlation witnesses (such as the Collins–Gisin–Linden–Massar–Popescu inequality for $d > 2$ [25]), and the smoothness-based robustness argument extends to each such witness. A detailed treatment of the $d > 2$ case is beyond the scope of this work but constitutes a natural direction for future research.

C. Single-shot corrections

The work bound (3) is an asymptotic (many-copy, or thermodynamic-limit) result. In the single-shot regime, the relevant quantities are the min- and max-entropies, and the corresponding free energies [11, 12]. The single-shot extractable work is

$$W_{\text{ext}}^{(1)} \leq k_B T [\ln 2 - H_{\min}(A|E)_\rho], \quad (21)$$

where $H_{\min}(A|E)_\rho$ is the conditional min-entropy (in nats, consistent with the natural-logarithm convention used throughout this work). The smoothness argument for the robustness hierarchy remains applicable, since $H_{\min}$ is a continuous function of the state, and its dependence on θ is inherited from the smoothness of $E(\theta)$. However, the quantitative values of the aggregate robustness parameter $\mathcal{S}_W$ would be modified, and the three-model hierarchy, while expected to persist qualitatively, deserves independent verification in the single-shot setting.

D. Connection to device-independent protocols

An analogous robustness advantage of quantum (and hypothetical beyond-quantum) correlations appears in device-independent quantum key distribution [26] and certified randomness expansion [27], where violation of a Bell inequality certifies the security of a protocol even when the measurement devices are imperfect. The thermodynamic robustness identified here—quantum correlations degrade more gracefully under misalignment—is a manifestation of the same structural feature: the smoothness of quantum correlations at their extrema provides intrinsic noise tolerance that is absent in classical theories and maximally present in no-signaling theories.

VIII. DISCUSSION AND RELATION TO PRIOR WORK

Our results complement and extend several strands of the literature on correlations as thermodynamic resources.

The foundational connection between correlations and work was established by Oppenheim *et al.* [9], who introduced the “work deficit”—the gap between globally and locally extractable work—as a measure of quantum correlations. While the work deficit quantifies how much

correlation *is* present, our analysis addresses a different question: how *robust* a given correlation is as a fuel. The two perspectives are complementary.

Zurek [8] connected quantum discord to the work extractable by Maxwell’s demon, showing that the “quantumness” of correlations has thermodynamic consequences. Our work extends this perspective beyond quantum mechanics to include hypothetical no-signaling correlations, and it identifies robustness—rather than the absolute amount of extractable work—as the key distinguishing feature.

Perarnau-Llobet *et al.* [28] provided a comprehensive framework for extractable work from general correlated states, proving tight bounds for local and global extraction protocols. Our work specializes to the Bell-experiment scenario and reveals a hierarchy that is not visible in bounds based solely on the amount of correlation.

Funo *et al.* [29] showed that entanglement can provide a thermodynamic advantage in work extraction under global operations, while Hovhannisyan *et al.* [30] demonstrated that entanglement generation is not *necessary* for optimal work extraction by local operations. Our finding that $W_{\max} = k_B T \ln 2$ for all three correlation classes is consistent with the latter result: the peak work is limited by the communicated information, not the nonlocality of the correlations. The advantage of nonlocality lies instead in the robustness with which this peak can be approached.

Gallego *et al.* [31] studied operational advantages of nonlocal correlations in information-theoretic tasks. Our work provides a thermodynamic analog of such advantages: the operational advantage is not a higher yield but a greater tolerance to imprecision.

The resource-theoretic approach to thermodynamics [10–12] introduces families of generalized free energies and establishes “second laws” beyond the standard one. Our bound in Eq. (3) is the asymptotic (many-copy) version of these constraints; the single-shot corrections (see Sec. VII C) would modify the quantitative values of $\mathcal{S}_W$ but not the qualitative robustness hierarchy, which is determined by the smoothness of $E(\theta)$.

It is important to clarify the scope of our results. The bound in Eq. (3) is standard within information thermodynamics: for systems coupled to a heat bath at temperature T , under the usual assumptions of the generalized second law, the maximum average work extractable in a cyclic protocol is bounded by $k_B T$ times the (appropriate) mutual information. In our operational setting we apply this to the classical outcome registers generated by the parties’ measurements. The expression for I_θ in Eq. (7) is similarly model-independent: it follows solely from assuming binary outcomes and uniform marginals, together with the correlator $E(\theta) = \langle AE \rangle_\theta$. What is model-dependent is the specific functional form of $E(\theta)$: the linear function is a standard (and, within a natural isotropic LHV comparison class, extremal) rotationally covariant local model; the cosine arises from the singlet

state; and the step function is an idealized no-signaling form reaching $\mathcal{S}_{\text{CHSH}} = 4$ [24]. However, as argued in Sec. VB, the robustness hierarchy depends only on the leading-order scaling of $E(\theta)$ at its extrema and is therefore generic within each theory. We emphasize that the robustness comparison is made within the class of rotationally covariant correlations satisfying $E(0) = -1$; outside this class (e.g., constant classical correlations that ignore the angle) the hierarchy does not apply.

IX. CONCLUSION

Correlations between a system and its environment are a thermodynamic fuel, whose value in our binary-outcome, uniform-marginal setting is limited by their mutual information. A single perfectly correlated bit can yield at most $k_B T \ln 2$ of work, as exemplified by a Szilard engine powered by shared information [4]. The hierarchy among classical, quantum, and stronger-than-quantum correlations lies not in this peak value but in the robustness with which it can be accessed. Quantum correlations provide a distinct operational advantage over classical ones through their resilience to misalignment, a feature directly linked to the smoothness of

their correlation function at its extremum. Engines consuming correlations do not violate the second law; they demonstrate that information must be treated as a physical resource [5, 6]. The robustness hierarchy—classical $<$ quantum $<$ no-signaling—provides a new thermodynamic signature that parallels the CHSH correlation hierarchy, and suggests that the “cost” of stronger correlations is not energetic but metrological: classical strategies demand precise alignment, quantum strategies are more forgiving, and no-signaling strategies are maximally tolerant. This hierarchy is established within the natural comparison class of rotationally covariant correlations satisfying the singlet boundary condition.

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